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**ACADEMIC PERFORMANCE OF BACHELOR IN SECONDARY
EDUCATION IN COLLEGE ALGEBRA**

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**A PAPER PRESENTED TO THE FACULTY OF COLLEGE
OF TEACHER EDUCATION KING'S COLLEGE OF THE
PHILIPPINES PICO, LA TRINIDAD, BENGUET
IN PARTIAL FULFILMENT TO
THE DEGREE**

BACHELOR OF SECONDARY EDUCATION

December 2015

APPROVAL SHEET

This study entitled "Academic Performance of the Bachelor of Secondary Education in College Algebra" prepared and submitted by Fei Y. Baguitan, Jenny P. Kimbes, Claire M. Marcos, and Jareth G. Pal-iwen in partial fulfilment to the requirements of the Degree BACHELOR OF SCIENCE IN SECONDARY EDUCATION has been examined and is hereby recommended for acceptance and approval for final defense.

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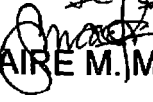
CERTIFICATION

This is to certify that all sources in this dissertation have been properly acknowledged and duly cited.

This is to certify further that this research is an original undertaking and has not been submitted for another degree.

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ACKNOWLEDGEMENT

The researchers would like to express their deepest and profound gratitude to their adviser Christine Flor B. Wansi, for her untiring support and professional guidance in the preparation and completion of the study.

The researcher also wishes to extend their profound gratitude and sincerest appreciation to the people who helped them in the conceptualization, preparation and completion of this work:

Dr. Ramon D. Santiago, class instructor for his undying patience and for guiding the researchers in the construction of the study.

Mr. Ruben M. Gambulao Jr. for the intelligent ideas and valuable suggestions in the construction of the questionnaires;

Dr. Rufina D. Cadsí, Dean of College of Teacher Education for allowing the researchers to conduct the study and to the Instructors who parted their time and allow the researchers to conduct the study during their class hour;

Dr. Jovita Fernandez, Head of Registrar for being kind and considerably assisted the researchers in getting the final grades of the respondents.

Ma. Jesusa L. Gambulao the editor of this research.

To the member of the panelist Ms. Rose Ann P. Fernandez, Dr. Ramon D. Santiago, and Dr. Rufina D. Cadsí the chairman for the refinement of the study.

To the Bachelor of Secondary Education junior students who served as the respondents for patiently and willingly answering the check-list questionnaire needed in this research work;

Their parents, siblings, relatives and friends who assisted in finding sources, made clarifications and provided financial and moral support;

And to the Almighty God who gave them the wisdom, knowledge, and understanding. He is the inspiration and guide to accomplish and to complete the study with patience and perseverance. The most genuine thanks are for all the blessings and incalculable delightful compassion as well as His chance to extend and sustain utmost gratitude and acknowledgment.

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ABSTRACT

This study was conducted primarily to determine the academic performance of Bachelor of Secondary Education students in College Algebra. SY 2015-2016. Specifically, it sought to answer the questions, What is the level of the academic performances of the respondents in College Algebra; What is the level of effect of the factors to the academic performances of the Bachelor of Secondary Education in College Algebra?

This study discovered that the academic performance of the Bachelor of Secondary Education is satisfactory with a weighted mean of 80.3. The factors affecting the academic performance of the respondents includes the student factor, teacher factor, and classroom setting environment factor and most of these factors were affecting much the performances of the respondents. The learning performance of the students in College Algebra is affected much by the student factors as it is implicated by its total weighted mean 2.56. Learning style ranks first with an average weighted mean of 2.68 and the least with an average weighted mean of 2.33 stated to be affecting moderately the performance of the respondent.

The teacher factors gathered a higher result shown by its total weighted mean 2.84 compared to the other factors affecting the learning performances of the students in Mathematics. Moreover, it affects much the Academic performance of the students in college Algebra. The students gave priority to the mastery of subject matter having a weighted mean of 3.22. On the other

hand the well-groomed and well poised with an average weighted mean of 2.37 who ranked 6 affects moderately their academic performances.

The classroom setting environment is said to be affecting moderately the Academic performance of the BSEd with a total weighted mean of 2.49. Ventilation and classroom capacity both ranked 1.5 with an average weighted mean of 2.73 affects moderately. Seating arrangement ranks last with an average weighted mean of 2.32 is said to be affecting moderately the respondents' performance.

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INTRODUCTION

This chapter presents the background of the study, conceptual frameworks, statement of the problem, scope and delimitation of the study, and significance of the study

Background of the Study

Mathematics is the study of topics such as quantity numbers, structure, space, and change. College Algebra the branch of mathematics that deals with general statements of relations, utilizing letters and other symbols to represent specific sets of numbers, values, vectors, etc., in the description of such relations. It serves as a gatekeeper for the higher mathematics plays a vital role in people's daily lives as cited by Salazar (2001) "that citizens of the modern world could not afford to be ignorant of Mathematics because the world is highly mathematical. Hence, effective mathematical instruction has become an absolute necessity in all levels of education. Furthermore Lubina (2004) cited that mathematics plays an important role in many aspects of life. It helps to understand and use quantities like weight, height and age properly. It helps individual to determine the size of population and predict whether country's available resources are enough to meet the needs of the people. It has become a useful with the emergence of computer and other technological devices. Indeed there is a need to learn mathematics in order to function effectively and become active participants in this changing information —based society. Villanueva also (2009) stated that mathematics is inseparable to man's life as it is a solid foundation satisfying his wants. He further mentioned

that through mathematical concepts and ideas, he may be able to appreciate the beauty of life, the environment and the real world.

Worldwide, there have been longing and urge from the early centuries to improve the mathematical performances of every students. The Third International Mathematics and Science Study studied the performance of students among 41 nations to determine the level of their performances. In the United States, there have been trends in mathematics performance in early 1990's. The National Assessment of Education Progress also known as "The Report Card" reported that student's performance in mathematics improved although not at a consistent rate across three decades because there have been decline occurred (Campbell et. al, 2000).

In our country in 2011, Science Secretary Mario Montejo wanted to create a viable math courseware and test whether it could take an impact on students learning together with Science Undersecretary Fortunato de la Pena and Education Undersecretary Yolanda Quijino chaired the project "Technology Package for Students Learning Empowerment" worked to make a courseware neither too fast nor distracting, providing sufficient time for the students to understand math concepts. In 2001 also the Science Education Institute and the Department of Science and Technology financed and monitored a project wherein the Department of Education made Possible the pilot testing of the material in public schools supported by a lot of institution to further improve the performance in mathematics.

The Level of College Algebra education has long been recognized as a major factor in development but an alarming observation of the student's performance reveals that they excel in knowledge acquisition but fare considerably low in lessons requiring higher order thinking skills. Thus, how to measure student performances must be the top priority in this area of concerned educationist. As cited by Benito in his study of Cabatay in 2011 "that mathematics is learned because of many reasons. Firstly the mastery of basic mathematics skills is needed in order to cope with the demand of life. Such demands include being numerically literate, gaining tools for future development, developing the prerequisite for further education, and appreciating the relationship between mathematics and technology. Secondly, mathematics is the language of the sciences, and many disciplines depend on this subject as a symbolic means of communication. Thirdly, mathematics education can play an important part in developing student's general decision making and problem solving skills.

But every year many students struggle with College Algebra not only in high school but also in college, all too many students are expected to fail. Shirly Bagwell a long time Tennessee teacher warns that "to expect all students to master algebra will cause more students to drop out." But a definitive analysis by the Georgetown Center on Education and the Workforce forecast that in the decade ahead a mere 5 percent of entry-level workers will need to be proficient in Algebra or above as cited by Credit Adam Hayes. Peter Braunfeld of the University of Illinois tells that our civilization would

collapse without mathematics. He's absolutely right. Algebra is the mystical portal to prosperity, as Duncan pointed out in his speech in 2011, that students who finish college algebra are twice likely to earn a college degree. It is true that more advanced math is necessary to understand every facet of advanced topics.

Academic performance refers to how they cope with or accomplish different tasks given to them by their teachers. It is the ability to study and remember facts and the ability to communicate acquired knowledge verbally or down on a paper, which are commonly measured by means of grades. Academic performance, which is measured by the examinations resulted is one of the major goals of schools. Schools are established with the aim of imparting knowledge and skills to those who go through them and behind all this is the idea of enhancing good academic performance.

The low passing rates for college algebra together with lots of bashers such: Why do I need to take college algebra? Why should I learn college algebra? When would I ever use algebra in real life? What's the point of learning algebra? Is it necessary? I'm only taking this because the university makes me. But I wouldn't need this stuff for my job. When will I, to a certainty, need everything thought in algebra?" These are some questions and statements addressed by the students. A lot of students can't really see the essence of learning mathematics in their life despite explaining the importance of mathematics the students still have the negative attitudes towards the subject. Come to think of it, why do students have common ideas regarding

math? Why is it they all see math hard and difficult subject? Indeed didn't discourage rather it ignites the researcher's imagination and curiosity to determine the performances of the Bachelor of Secondary Education juniors in College Algebra and the factors affecting their performances together with the effect in their learning.

So instead of thinking about the answer to the question it is easier to let them see the mathematical sequoia, to tell them how great it is and think how can we better prepare to study algebra? It would surely not be easy but it is worth doing.

Conceptual Framework

Socio-economic status is determined to be a predictor of mathematics achievement. Studies repeatedly discovered that the parents' annual level of income is correlated with students' math achievement scores (Eamon, 2005) Socio-economic status was found significant in primary math and science achievement scores (Ma & Klinger, 2000).

A number of studies showed that parents with higher socio-economic status are more involved in their children's education than parents of lower socio-economic status. This greater involvement results in development of positive attitudes of children toward school, classes, and enhancement of academic achievement (Epstein, 1987). It is believed that low socio-economic status negatively influences academic achievement, in part, because it prevents students from accessing various educational materials and resources, and creates a distressing atmosphere at home such as possible

disruptions in parenting or increased likelihood family conflicts (Jeynes, 2002). For these reasons, socio-economic status of a student is a common factor that determines academic achievement.

Being successful in math involves the ability to understanding one's current state of knowledge, build on it, improve it, and make changes or decisions in the face of conflicts. To do this requires problem solving, abstracting, inventing, and proving (Romberg, 1983). These are fundamental cognitive operations that students need to develop and use it in math classes. Therefore, instructional strategies and methods that provide students with learning situations where they can develop and apply higher-order operations are critical for mathematics achievement.

In the literature, it is pointed out that for students to accomplish learning, teachers should provide meaningful and authentic learning activities to enable students to construct their understanding and knowledge of this subject domain (Wilson, 1996). In addition, it is emphasized that instructional strategies where students actively participate in their own learning is critical for success (Bloom, B. 1976). Instructional strategies shape the progress of students' learning and accomplishment.

Many studies report that what teachers know and believe about mathematics is directly connected to their instructional choices and procedures (Brophy, 1990). Geliert (1999) also reported that "in mathematics education research, it seems to be undisputed that the teacher's philosophy of mathematics has a significant influence on the structure of mathematics

classes". Teachers need to have skills and knowledge to apply their philosophy of teaching and instructional decisions.

Teachers not only need knowledge of a particular subject matter but also need to have pedagogical knowledge and knowledge of their students (Bransford et al. 2000). Teacher competency in these areas is closely linked to student thinking, understanding and learning in math education. There is no doubt that student achievement in math education requires teachers to have a firm understanding of the subject domain and the epistemology that guides math education (Ball, 1993) as well as an equally meticulous understanding of different kinds of instructional activities that promote student achievement. Competent math teachers provide a roadmap to guide students to an organized understanding of mathematical concepts, to reflective learning, to critical thinking, and ultimately to mathematical achievement.

School context and its facilities could be an important factor in student achievement. In fact, identifying factors related to the school environment has become a research focus among educational practitioners. For instance, research suggests that student achievement is associated with a safe and orderly school climate (Reynolds et al. 1996). Researchers also found a negative impact on student achievement where deficiencies of school features or components such as temperature, lighting, and age exist. In a study by Harner (1974), temperatures above 23° C adversely affected mathematics skills. In terms of the condition of school building, Cash (1993) found student achievement scores in standard buildings to be lower than the scores of

students in above standard buildings. In addition, Rivera-Batiz and Marti (1995) conducted multiple regression statistical analysis to examine the relationship between overcrowded school buildings and student achievement. The findings indicated that a high population of students had a negative effect on student achievement.

Students' attitude towards mathematics influences the efforts they put in understanding and practicing mathematical concepts and skills. According to the National Research Council (2000) as cited in Akey (2006), Students' beliefs about their competence and their expectations for success in school have been directly linked to their levels of engagement, as well as to emotional states that promote or interfere with their ability to be academically successful. Thus attitudes determine the effort a student is likely to put in his learning of the subject mathematics. It is therefore necessary for mathematics teachers to strive and sustain positive attitudes towards mathematics for good performance in the upper classes (Benson, 1999).

Students' educational outcome and academic success is greatly influenced by the type of school which they attend. The school one attends is the institutional environment that sets the parameters of a students' learning experience. Depending on the environment, a school can either open or close the doors that lead to academic achievement. According to Considine (2002) the type of school a child attends influences educational outcomes. Considine (2002) cite Sparkles (1999) whose study in Britain shows that schools have an independent effect on student attainment and that school effect is likely to

operate through variation in quality and attitudes, so teachers in disadvantaged schools often hold low expectations of their students which compound the low expectations the students have, hence leading to poor performance by the students. Kwesiga (2002) agrees that school has an effect on the academic performance of students but argued that school facilities determine the quality of the school, which in turn influences the achievements, and attainment of its pupils. Sentamu (2003) argues that schools influence learning in the way content is organized and in the teaching, learning and assessment procedures. All these scholars agree in principle that schools do affect academic performance of students.

Effective Schools Model by Lezotte (2010), according to this model, an effective school is a school that can measured student achievement terms, demonstrates the joint presence of quality and equity. According to Lezotte (2010), strong instructional leaders are proactive and seek help in building team leadership and a culture conducive to learning and professional growth. A safe and orderly school is defined as a school climate and culture characterized by reasonable expectations for behavior, consistent and fair application of rules and regulations, and caring, responsive relationships among adults and students (Lezotte, 2010). Classrooms are warm and inviting, and learning activities are purposeful, engaging, and significant. Personalized learning environments are created to increase positive relationships among students and between students and their teachers. Teaching advanced skills and teaching for understanding together with basic

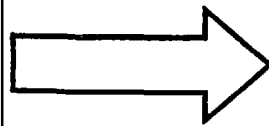
skills are required for all students to achieve at high levels. Frequent monitoring of teaching and learning requires paying attention both to student learning results and to the effectiveness of school and classroom procedures (Lezotte, 2010). Teaching is monitored by teachers themselves through self-reflection and by supervisors for program and teacher evaluation. Opportunity to learn and student time on a task simply means that students tend to learn most of the lessons they spend time on. Time on task implies that each of the teachers in the school has a clear understanding of what the essential learner objectives are, grade-by-grade and subject-by-subject. Once it is clear what students should be learning, they should be given time to learn it. In an effective school, teachers allocate a significant amount of classroom time to instruction on the essential skills. Students of all abilities, races, gender, and socioeconomic status have equal opportunities to learn (Lezotte, 2010).

The independent variables are the academic performance of the students and factors affecting academic performance. And for the dependent variable are the level of academic performance and degree of effects of factors to the academic performance. This factors are teacher factor such as well groom and well poised, verbal and non-verbal mannerism, choice of instruction materials, clarity of voice, teaching strategies and mastery of subject matter, student factor such as mathematical ability, learning style, economic status, health condition, aptitude and number of units enrolled and classroom setting factor such as ventilation, seating arrangement, classmates, cleanliness, and classroom capacity. See figure 1. Paradigm of the study.

Paradigm of the Study

Independent Variable

- Academic performance of the students
- Factors affecting the students.



Dependent Variable

- Level of learning performances
- Degree of effects of the factors to academic performances

Figure1 Paradigm showing the interrelationship of variables used in the study.

Statement of the Problems

This study was undertaken primarily to determine the learning performance of the College of Teacher Education Bachelor in Secondary Education students' in College Algebra First Semester.

Specially, it sought to answer the following questions:

1. What is the level of Academic Performances of Bachelor in Secondary Education Junior's in College Algebra?
2. What is the level of effect of the factors to academic performance of the Bachelor of Secondary Education Students in College Algebra?
 - a) student
 - b) teacher
 - c) classroom setting

Scope and Delimitation of the Study

The study was conducted at King's College of the Philippines located at Pico road, La Trinidad, Benguet. The study focused on the Academic Performance of Bachelor of Secondary Education students in College Algebra. The subject of the study was limited to the third year students who had already took up their College Algebra subject.

Significance of the Study

There have been much of discussions over the years about the usefulness of study of College Algebra. Everyone seems to have a different viewpoint on the issue. Some believe that College Algebra has a little use in

the working world and so it is not a subject that should be taught at a higher level in secondary school. Others argue that College Algebra does serve a profound purpose that knowledge of College Algebra is absolutely necessary. It has an important bearing on the intellect as such. It promotes the habit of accuracy and exactness. It develops the reasoning ability of a man as well as it increases mental alertness, thus should be taught. It is essential discipline that needs to be augmented in education to equip students with skills necessary for achieving in higher education, career aspiration, and personal fulfillment.

The outcome of this study shall then act to the researchers as a basis for further improvement of how to deliver their lesson and subject matter when they will be practicing their chosen profession.

The study will serve and act as a diagnostic tool to guide the teachers, coordinators and education planner in designing the curriculum that suits the needs of the learners in the subject area. It leads to the improvement of curriculum, of methods used to suitable learning strategies of the students and in constructing meaningful strategies that will eliminate the negative impact of the subject to the learners.

To the respondent they will be able to evaluate the factors affecting their performances and be aware on how they would cope and deal with it to lessen the impact and burden.

METHODOLOGY

This chapter presents the method of research used, the population frame and sample size, instrumentation, the data gathering procedure, and the statistical treatment of the data used in the study.

Research Design

This study made use of descriptive research method which is designed with test questions and grades of the students as the main instruments for gathering data. According to Santiago (1985) descriptive research concerns the present situation, prevailing conditions, current practices, contemporary events, characteristics of group individuals, their behavioral patterns and attitudes. A study designed to depict the participants in an accurate way. Hence, this method was used to analyze the opinions or perceptions of the students on the factors affecting learning performance of Bachelor of Secondary Education third year student in math at King's College of the Philippines.

Answer to the first problem was dealt with by identifying the level of performance of the third year Bachelor of Secondary Education students by taking their grades at the Office of the School Registrar.

Answer to the second problem was dealt with by identifying the level of effect of the factors affecting the performance of Bachelor of Secondary Education students.

Population and Locale of the Study

The respondents of the study are the currently enrolled 3rd year students of King's College of the Philippines S-Y 2015-2016. They were selected using purposive sampling and since there are only five majors in the Bachelor of secondary Education all of them are the included. The population consists of 16 math majors, 7 Filipino majors, 7 General Science majors, 33 English majors, and 27 MAPEH majors.

Data Gathering Instrument

The instruments used are questionnaires, which are a necessity for a descriptive method. The contents of the questionnaire were constructed with the aid of all information gathered with respect to the curriculum requirement, related studies and questions covering the factors affecting the performances of the third year student.

By all means, the researcher designed the questionnaire in proper form to accommodate sufficient and relevant information required from the respondents.

The questionnaire is divided into two parts. The first part concentrated on the grade of the respondents. Weighted mean was used to modify the level of the grades of the respondents considering the seven scales. Respondents are requested to check the space provided for each option.

The instrument is valid in the sense that the data was taken from the Office of the School Registrar and the questionnaire was constructed by the researchers and then submitted to the adviser. With the help of the adviser and the critiquing, suggestions, and comments of some math teachers the questionnaire was formulated. The suggestions and comments were then incorporated in the final drafting of the survey questionnaires, after which the instruments were subjected to evaluation and approval before their reproduction and administration to all respondents.

Data Collection Procedures

Immediately after the reproduction of the questionnaires the researchers first sought the permission from the college dean to float the questionnaires to the respondent and the permission from the instructor of the class. For the performance of the respondents the researchers sought the permission from the good office of the college dean and then submitted to the office of the registrar asking for the grades of the respondents when they took up College Algebra.

The researchers spent great time in developing their questionnaire so as to serve its intended respondents. With regards to the data gathering they first sought the permission from the College Dean to float the questionnaire to the respondents, followed by permission from the instructors holding their classes.

The performance of the respondents was based from the grades of the students in College Algebra, which was retrieved from the School Registrar's Office after the permission was signed by the office of the College Dean.

Treatment of the Data

Having retrieved the accomplished data and checklist questionnaire, the researcher compiled and tabulated the data. The researchers used the weighted mean to measure the level of performance and the degree of the factors affecting the learning performance of the third year Bachelor of Secondary Education students.

The 4-point scale was used for the level of effect of the Factors to Academic Performance of the Bachelor of Secondary Education.

Numerical Equivalent	Statistical Limits	Descriptive Equivalent
4	3.26-4.00	Very much affects
3	2.51-3.25	Much affects
2	1.76-2.50	Moderately affects
1	1.00-1.75	Less affects

The 7-point scale was used for the Level of Academic Performance of Bachelor of Secondary Education.

Numerical Equivalent	Statistical Limits	Descriptive Equivalent
7	96 —100	Outstanding
6	93 — 95	Excellent
5	90 - 92	Superior
4	78 — 83	Very Satisfactory
3	75 - 77	Satisfactory
2	70 - 74	Fair
1	84 — 89	Poor

The following formula was used in the computation of weighted mean.

$\mu = \frac{\sum xE}{N}$ Where:
E= summation

f= frequency
x= weight of each scope
N= number of samples

RESULTS AND DISCUSSION

This chapter summarized the gathered data, assembled and presented in tabular form for further analysis and interpretation. Essentially, the discussion is focused on the findings on the specific problems asserted in Chapter 1.

Level of Performance of Bachelor of Secondary Education in College Algebra

Table 1 shows the level of performance of College of Teacher Education students in College Algebra.

As shown in the Table 1 above, Bachelor of Secondary Education junior's performance in College Algebra with a weighted mean of. 80.3 interpreted to be satisfactory. The weighted mean below revealed that the respondent from Bachelor of Secondary Education has attained the minimum required knowledge with regard to the subject. But then they are expected to get hard up for the higher mathematics because what they have acquired just the minimum.

Table 1. Level of Academic Performances of Bachelor in Secondary Education respondents in College Algebra.

Scale	Frequency	De	rank
96-100	0	outstanding	7
93-95	3	Excellent	6
90-92	9	Superior	5
84-89	16	Very satisfactory	3
78-83	20	satisfactory	2
75-77	27	Fair	1
70-74	15	Poor	4
Total	90		
Average Mean		80.3 (satisfactory)	

In the study conducted by Lambitco, et al. the findings of the study revealed that the student means grades in college algebra was satisfactory. More studies were conducted in this area and researchers come up with their recommendations as well. But still the problem of low performance in college algebra appears to be pertinent in all levels of school education. This of course is a serious issue to be tackled possibly by the teachers concerned themselves since they ranked first to be affecting the performances of the students. Thus, teacher related factors such as mastery of subject matter, method of teaching, teaching strategies, and personality of teacher must be the top priority in a class room to create interest among the students. Since mathematics is regarded by the students as a hard subject and is abstract in nature, the students found it difficult to obtain a very high grade in it. Their foundation in high school mathematics must be really good in order to be ready for college work. The students' mathematical ability must be high in order to obtain good performance in college algebra because learning mathematics is directly related to mathematical ability. The higher the level of Mathematics anxiety of the student, the lower is his/her performance in mathematics.

Other researchers also considered other variables significant to the performance of students in math. Some of these researches included student ability, teacher factors, mathematics proficiency and English language among others. In the descriptive correlation study of the students perception towards Mathematics and their impact in Algebra subject, Derano, et al. tried to find out

if there exist a significant relationship between teacher factors and students' Algebra proficiency and the relationship between students' study habits and attitude in Math and their Algebra proficiency. After completing the study, researchers found out that teacher's strategies, methodologies, competency and motivation towards Algebra is not significantly correlated. However, a significant direct relationship existed between students' study habits, preparation in Math and their Algebra proficiency.

Students each possess their own preferred learning style. So, all classes are comprised of this accumulation of individual students each with a different and unique learning style of preference. The challenge for the teacher is to choose their approach to teaching the class. Teachers have at least three paths they can choose; they can choose to create and follow one lesson for a group of students, construct individual lesson plans utilizing the preferred learning style of each of the students, use a method that includes individual students within one group lesson, or some other choice. A method that has enough relevant information or interest level to help a student to relate new material to information they already know would help a student to remember and work with new information. It would enable transference of knowledge. If information is related to student interests or goals, the student may be motivated to seek out more related information. Students may want to make certain they learn the necessary knowledge or skills to meet the threshold of the outcome standards intended skills/ knowledge to meet the standards of the subject matter. Regardless of the choice the teacher attempts to engage the

student, the teacher must be able to motivate the student to pursue the knowledge acquisition whether it is simply the quest for the necessary knowledge, the individualization, or the enabling of association and transference in order for the outcome standards to be met. The more we as educators address teaching techniques, seeking a solution to meet the variations of learning/teaching styles, the greater the likelihood that we will reach the students with a strong enough portion of the instruction lesson to impart the vital parts of necessary information to the students in a format that they can understand to be deemed successful. The resultant effect of such a match is that students will have better attitudes, learn more material, and the outcome of the class as a whole will meet a higher.

College algebra has several benefits. It exercises the mind and prepares the student to learn and better understand difficult concepts. It increases the reasoning and understanding skills of people. G.H. Hardy (1940) identified six traits that Mathematicians display. The list of the traits of a mathematician include: (a) intellectual curiosity; (b) a creator of patterns; (c) seeks connections between mathematical ideas; (d) seeks mathematical accuracy; (e) generates mathematical generalizations; and (f) seeks mathematical efficiency and economy. Hardy's traits were intended to describe the practicing mathematician. Polya, et al. (as cited in Singh, 1997) also made contributions to the discussion about the kinds of habits displayed by a mathematician. College algebra is also beneficial to other goals and objectives of many of the math curriculum and professional career development paths,

such as using mathematical modeling to make predictions and to solve problems. Learning how to interpret results of algebraic calculations is not highly dependent on the ability to perform the calculations themselves. The procedural understanding is influenced by the careful attention to the sequential aspect of the process. Outcomes indicated that even a curriculum in which emphasis was found on mathematical modeling in real-world contexts for using the algebraic calculations does not necessarily produce students who have mastered that ability (Huntley et. al, 2000).

**Factors Affecting the Level of Academic Performances
of Bachelor in Secondary Education
in College Algebra**

The performance of the students in academics is not only influenced by their own characteristics gifted by the nature but also various factors are involved in these achievements. For the economic and social development of the society, it is necessary to provide our children with the quality education. In recent years, most of the efforts have been made to search out the factors that can affect the student academic performances. Students' academic gain and learning performance is affected by numerous factor including gender, age, teaching faculty, students schooling, father/guardian social economic status, residential area of students, medium of instructions in schools, tuition trend, daily study hour and accommodation as hostelries or day scholar. Many researchers conducted detailed studies about the factors contributing student performance at different study levels.

Table 2 below shows the factors affecting the academic performance of the Bachelor of Secondary Education in College Algebra.

Table 2. Factors Affecting the Performance of Bachelor of Secondary Education in College Algebra.

Factors	WM	DE	rank
1. Student factor	2.56	MuA	2
2. Teacher factor	2.84	MuA	1
3. Classroom Setting factor	2.49	MoA	3
Average Mean	2.63	MuA	

In the table above, teacher factor ranked first to be affecting the level of performances of the students with a total average weighted mean of 2.84 interpreted • to be much affecting the performance of the respondents. Many factors contribute to a student's academic performance, including individual characteristics and family and neighborhood experiences. But research suggests that, among school-related factors, teachers matter most. When it comes to student performance on reading and math tests, a teacher is estimated to have two to three times the impact of any other school factor, including services, facilities, and even leadership. Mary Chamberlin (2002) said that great teachers make a difference. They have the passion that seeps through the skin-a love of learning. Huang & Moon (2009) documents that teacher qualification account for approximately 40 to 60 percent of the variance in average of students' achievement in assessment. Maundu (1986) concludes also that there was

significant correlation between teacher qualification and pupil performance in Kenya. Recent evidence have suggests that a teacher's impact on student achievement remains reasonably consistent even if the teacher changes schools and regardless of whether the new school is more or less advantaged than the. old one. Research had also showed that effective teachers are the most important factor contributing to student achievement. Although curricula, reduced class size, district funding, family and community involvement all contribute to school improvement and student achievement, the most influential factor is the teacher. Choosing effective teachers is critically important for schools trying to improve their performance.

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Student factor ranked second to be affecting the level of performance of the students with a total weighted mean of 2.56 interpreted to be much affecting the learning performances of the respondents. Noble (2006), students' academic accomplishments and activities, perceptions of their coping strategies and positive attributions, and background characteristics (i.e., family income, parents' level of education, guidance from parents and number of negative situations in the home) were indirectly related to their composite scores, through academic achievement. The students face a lot of problems in developing positive study attitudes and study habits. Guidance is of the factor through which a student can improve his study attitudes and study habits and is directly proportional to academic achievement. Hijaz, et. al., (2006) observed that there is a negative relationship between the family income and students' performance.

The third factor that affects level of performances of the respondents is the classroom factor with a total weighted mean of 2.49 affects moderately. As cited by Marshall (2012) in her study that the physical environment is one of the easiest to understand. The influences are obvious in nature and comprise such things as the lighting, flooring, furniture, and architecture and can also encompass layout and spaces, accessibility and cleanliness, as well as interior color schemes. In addition to the look of the material items, advising professionals should consider the feelings these items provoke. These physical spaces can convey nonverbal messages sometimes stronger than verbal by either welcoming or discouraging and valuing or disrespecting. The aggregate facet is founded on the fact that environments are transmitted through people and reflect the collective characteristics of the people who are influenced by them. These human characteristics influence the degree to which people are pleased or dissatisfied with their environment. These characteristics are made up of demographics such as gender, age, or racial-ethnic composition and psychological traits such as personality types, interests, and styles. The collective aggregate facet is composed of the staff and students that interact with the space.

Physical environment refers to physical characteristics of the room. The physical environment influences students the moment they arrive. The furniture, lighting, signage, comfort, and layout of the space impact how students perceive the environment. Physical classroom environment is a combination of different things i.e., lighting, temperature, ventilation system,

size of the room, floor, walls, desks, chairs, rugs, whiteboards, computers etc. Classroom management is a critical part of effective and successful instruction. Effective classroom management, which initiates with well-organized and efficient lesson planning preparation, helps a teacher to teach and students to learn. Students perform well in an optimistic classroom atmosphere and an environment in which they feel secure, safe, cared for and involved. From a student point of view, effective and successful classroom management provides students with opportunities to socialize while learning interesting content. From a teacher point of view, effective classroom management involves preventive discipline and interesting instruction (Lang and Hebert, 1995). To ensure a favorable classroom environment, it should be well equipped and facilitated. The classroom physical arrangement provides students with effective instruction and promotes smooth teaching learning process. Physical facilities should be provided as they are helpful in improving the overall performance of the school.

**Effects of the Factors in the Academic
Performance of Bachelor of Secondary
Education in College Algebra**

A) Student Factor Affecting the Academic Performance of Bachelor of Secondary Education Students.

Table 3 shows the results on the factors in the learning performance of bachelor of secondary Education Juniors respondents in student factor.

In student factors, the learning style ranked first, which can be classified and identified in many different ways. This means that Bachelor of secondary

Education perceived that individual's way of learning affects their learning performance in college algebra.

Generally, they are overall patterns that provide direction to learning and teaching. The term learning style refers to the concept that individuals differ in regard to what mode of instruction or study is most effective for them (Pashler, p. 105). According to Silver, Strong, and Perini (2000) the concept dates back to ancient Greek all the way to the Renaissance. They linked the learning style concept to Hippocrates "fourness" which when not in equilibrium cause persons to exhibit four types of personalities and William Blake's description of the four Zoas of human existence: the body and its senses; the heart and its capacity for love; the head and its ability to reason; and the spirit and its potential for creative imagination seem similar to that of Hippocrates. Silver et al. (2000) believed that evidence of the learning style concept can also be found in the spiritual stories of Indians of the North American Plains. The four human personality traits are given as wisdom, clarity of perception, introspection, and understanding one's emotions. Carl Jung (1923) cited in Silver et al. (2000) reclassified human "fourness" and advanced that humans use perception and judgment as cognitive functions to process information.

Table 3. *Student Factor Affecting the Academic Performance of the Bachelor of Secondary Education.

Student factors	WM	DE	Rank
1. Mathematical Ability	2.66	MuA	2
2. Learning Style	2.68	MuA	1
3. Economic status	2.58	MuA	4
4. Health condition	2.63	MuA	3
5. Aptitude	2.47	MoA	5
6. Number of units enrolled	2.33	MoA	6
Average mean	2.56	MuA	

Learning style also can be classified as a set of factors, behaviours and attitudes that facilitate learning for an individual in a given situation. Learning style ranked most because it can really affect student performance if the teacher technique or way in teaching is not suitable to every kind of learners specifically if they are a slow learner. They are not able to access information if their learning style is not suitable on the teacher way.

Riefler (2010) believe that the learning style theory is a myth. They agree that since differences among learners tend to affect their performance, they should be taken into account by teachers. They contend though that other factor such as learners' ability, background knowledge, and interest vary from person to person and when learning styles are emphasized these important elements are neglected in the analysis of their effect on learning.

Mathematical ability ranked second in student factors. Mathematical ability is the ability to obtain, process, and retain mathematical information (O'Donoghue, 2009) or as the capacity to learn and master new mathematical ideas and skills (Koshy et al. 2009). Mathematical ability can affect the performance of the learners because they may understand the topic or the lesson but when it comes to application they do not have the ability to obtain, and process the information or not able to apply it in a new situation.

Arithmetic ability could also be another predictor of math achievement. Arithmetic ability includes the skills such as manipulating mathematical knowledge and concepts in ways that transform their meaning and

implications. It allows students to interpret, analyze, synthesize, generalize, or hypothesize the facts and ideas of mathematics. Students with high arithmetic ability or mathematical reasoning can engage in tasks such as solving complex problems, discovering new meanings and understanding, and arriving at logical conclusions.

Arithmetic ability was determined by various studies as a critical factor on students' math achievement. For instance, in a study by Keeley (1993), arithmetic ability gave the highest correlation coefficient with mathematics achievement. Similarly, student achievement scores were found to be most strongly predicted by level of ability (Schiefele, 1995).

Health condition ranked third among the student factors. Student performance *can* be affected by their health condition because if they are not feeling well they cannot focus on their studies, it bothers them and if they are well they could perform better as Ross and Wu (1995) point out that education is associated with good health.

The fourth one is the economic status which also affects student performance because if their income is not sufficient it can really lessen their interest in going to school cause stress and worries. Amutabi (2003) discussed the impact of socioeconomic status on children's readiness for school:" The segregating nature of social class, ethnicity may well reduce the variety of enriching experiences thought to be prerequisite for creating readiness to learn among children. Social class, ethnicity, dictate neighborhood, housing,

and access to resources that affect enrichment or deprivation as well as the acquisition of specific value systems." As cited in the American International journal of Social sciences. APA (2001) described the relationship of family socioeconomic status to children's readiness for school, across all socioeconomic groups; parents face major challenges when it comes to providing optimal care and education for their children. For families in poverty these challenges can be formidable. Sometimes, when basic necessities are lacking, parents must place top priority on housing, food, clothing, and health care. Educational toys, games, and books may appear to be luxuries, and parents may not have the time, energy, or knowledge to find innovative and less-expensive ways to foster young children's development, Ominde (1964) observes that even in families with above average income parents often lack the time and energy to invest fully in their children's preparation for school, and they sometimes face a limited array of options for high-quality child care both before their children start school and during the early school years. Kindergarten teachers throughout the country report that children are increasingly arriving at school inadequately prepared to learn. According to Graetz (1995), one's educational success depends very strongly on social economic status of the parents. Considine and Zappala (2002) argue that families where the parents are advantaged socially, educationally and economically foster a high level of achievement in their children. The researcher agrees with Considine and Zappala (2002) because students from high social economic backgrounds are well exposed to scholastic

materials, which aid their intelligence. Sentamu (2003), argue that the type of school a child attends influences academic achievement. According to Minnesota measures (2007), a report on higher education performance, which was produced by the University of Minnesota, the most reliable predictor of student success in college is the academic preparation of students in high school. Social Economic Status (SES) according to Considine, et.al., (2002) is a person's overall social position to which attainments in both the social and economic domain contribute. They add that social economic status is determined by an individual's achievements in, education, employment, occupational status and income. In this study social economic status was characterized by family income, parental education and parental occupation. Graetz (1995) argues that children from high social economic status families perform much better at school compared to children from low SES families.

Family income, according to Escarce (2003) has a profound influence on the educational opportunities available to adolescents and on their chances of educational success. Escarce (2003) adds that due to residential stratification and segregation, low-income students usually attend schools with lower funding levels, have reduced achievement motivation and much higher risk of educational failure. When compared with their more affluent counterparts, low-income adolescents receive lower grades, earn lower scores on standardized test and are much more likely to drop out of school. Escarce (2003) is in agreement with Sentamu (2003) who argued that social class determines what school a child will attend and whether the child will pass the

examinations. Considine (2002) found that children from families with low income are more likely to exhibit the following patterns in terms of educational outcomes; have lower levels of literacy, innumeracy and comprehension, lower retention rates, exhibit higher levels of problematic school behavior, are more likely to have difficulties with their studies and display negative attitudes to school.

King (1989) used parents' occupation as a proxy for income to examine the relationship between income and achievement and found that children of farmers had fewer years of schooling than children of parents with white-collar jobs. They also determined that the schooling levels of both parents had a positive and statistically significant effect on the educational attainment of Peruvian children. They argue that how much education a child's parents have is probably the most important factor in determining the child's educational opportunities. They observe that the higher the attainment for parents, then the greater their aspirations for children.

The fifth one is the aptitude. Aptitude is the capability, ability, innate or acquired capacity for something and intelligence. Aptitude affects student performance because there are some people who are really gifted mathematically wherein they have the ability and the capability to understand math concepts. If you do not have mathematical skills and it will be hard to synthesize and analyse problems. And the one who ranked last is the number of units enrolled. This affects students' performance if they are not able to handle all the subjects that they enrolled. If they cannot balance their efforts it

will affect their level of their performance. There will be pressure and unbalanced effort.

B) Teacher Factor Affecting the Academic Performance of Bachelor of Secondary Education Students.

Table 4 below presents the result in the learning performances of the students in teacher factor.

Ranked first under the teacher's factor is the Mastery of subject matter with a weighted mean 3.22 interpreted as much affecting the respondent performances. It stated that mastery of subject matter is an essential component of teacher if teaching entails helping others learn, then understanding what is to be thought is a central requirement of learning myriad task of teaching such as selecting worthwhile learning activities, giving helpful explanations, asking for the productive questions, and evaluating students learning, all depend on the teachers understanding of what it is that students are to learn.

It is true that teachers' knowledge of the content they teach is a consistently strong predictor of student performance, even though studies differ in how strong its effects are. Research has also been consistent in

Table 4. Teacher Factor affecting the Academic Performance of Bachelor of Secondary Education.

Teacher Factors	VVM	DE	Rank
1. Well groomed and well poised	2.37	MoA	6
2. Verbal and nonverbal mannerism	2.64	MuA	5
3. Choice of Instructional Materials	2.90	MuA	3
4. Clarity of voice	2.76	MuA	4
5. Teaching strategies	3.16	MuA	
6. Mastery of Subject matter	3.22	MuA	
Average Mean	2.84	MuA	

finding positive correlations *between* years of teaching experience denotes mastery of the subject course and higher student achievement.

Teachers with more than five years in the classroom seem to be the most effective. Conversely, inexperience is shown to have a strong negative effect on student performance. Research consistently shows that teacher quality—whether measured by content knowledge, *experience*, training and credentials, or general intellectual skills—is strongly related to student achievement: Simply, skilled teachers produce better student results. Many researchers and analysts argue that the fact that poor and minority students are the least likely to have qualified teachers is itself a major contributor to the achievement gap.

Subject matter knowledge is widely acknowledged as a central component of what teachers need to know, research on teacher education has not, in the main, focus on the development of subject matter knowledge. Researchers specifically interested in how teachers develop a change have focused on other aspects of teaching and learn to teach for example, changes in teachers' role conceptions, their beliefs about their work, their knowledge of students, curriculum, or of a teaching strategies. Yet and known the teachers' development of subject matter knowledge seems to believe its importance in teaching and learning to teach.

The subject matter knowledge has impact combined set of qualifications measured by teachers licensure are only leading factors in teaching.

Ranked second is teaching strategies with weighted mean 3.16 interpreted as much affecting the respondent's performance. This indicates that selection of proper instructional strategy ensures the achievement of the stated instructional objective effectively. Instructional strategies are used in the presentation of lesson to help the students learn by ensuring the smooth delivery of the instruction. It is a process by which an instructional module, instructional phase or an entire course is delivered.

The primary purpose of teaching at any level of education is to bring a fundamental change in the learner (Tebabal, 2011). To facilitate the process of knowledge transmission, teachers should apply appropriate teaching methods that best suit specific objectives and level exit outcomes. In the traditional epoch, many teaching practitioners widely applied teacher-centered methods to impart knowledge to learners comparative to student-centered methods. Until today, questions about the effectiveness of teaching methods on student learning have consistently raised considerable interest in the thematic field of educational research (Hightower et al., 2011). Moreover, research on teaching and learning constantly endeavor to examine the extent to which different teaching methods enhance growth in student learning. But with active teaching strategies, the traditional lecture can be an effective lecture approach that it provides a way to communicate a large amount of information to listening to students. This *following can* help make the lecture approach effective (Cashin1990). Fit the lecture to the audience. Focus your topic — remember you cannot cover everything in one lecture. Prepare an outline that includes your

5.9 major points you want to cover in one lecture. Organize your points for clarity. *Select* appropriate examples or illustrations. Present more than one side of an *issue and be sensitive* to others perspective. Repeat points when necessary. Be aware of your audience, *notice* their feedback, be *enthusiastic*, you don't have to be an entertainer, *but* you should be excited by your topic.

Ranked *third* is choice of instructional materials with weighted mean 2.19 *interpreted* much affects. All instructional materials are aids to instruction. They do *not* replace *the* teacher. Choose instructional materials that best suits *in your instructional objectives*. Decide what *you* want to accomplish and then employ *all the* tools *that are* most *likely* to *achieve* results. Do not *let* media that is available to determine how or what you *will* teach. If possible, use variety of *tools*. *Using* videos, computers, overheads, and the chalkboard not only *keeps students' interest but also respond* to the needs of those receive information in different *ways*. Check out your instructional materials before class starts to be *sure it is working* properly. *Nothing is* more frustrating to you or to the student^{ts} *in* a process of instruction than to find out the overhead projector, for *instance does not work in the process of instruction*. For results, *abide by the* *general utilization guide on the use of media*. Learn how to use *the* *instructional materials*. Prepare *introductory* remarks, question or initial *comments you may need*. Provides a conducive learning environment for the *student and expect* *in the* objectives of lesson. Stress what to be watched or *listen to carefully*. *Then* state what they *will be expected* to do with the *information they will learn*.

Instructional materials are primary resources for English language arts also materials that are provided online as well as those generated by teachers. Instructional materials *including* textbooks, educational media, library media *print* and others.

Ranked fourth is clarity of voice with weighted mean 2.76 interpreted to be much affecting *the learning performances* of the respondents. As teacher we all know that *voice is a* teacher most valuable asset. The one thing teachers do most often is talk rightly or *wrong*. They talk *for long* periods each day, *for many years*. They rely on their voices to *great* extent, and with it they transmitted *not only information*, but also mode, atmosphere and emotions.

Moreover, how they speak and what are voice sounds like a crucial impact classes. Harmer (2007) recommends three issues we should think about when considering the use of the in management of teaching audibility, variety in conversation. The teacher's voice need to be audible. Students at the back of the class must be able to hear then as well as tools at the front. Switching from one-to-one and small groups to whole class in some situations teachers need to have adaptability as one of their abilities. Audibility, however, can divorce from voice quality. Teachers need a voice which projects well and appropriately in all these circumstances and which is also pleasant to listen to. Teachers do not need to shout to be heard or make it understood. Good voice projection is more important than volume. Both speaking too soft or too loud are irritating and unhelpful to the students. Voice problems can be disguised

as other physical and mental. Teachers program *has been* develop to assist teaching.

Ranked fifth is *verbal and non-verbal mannerism with weighted mean 2.64* interpreted *much affect. Establishing* a good communication between student and the teacher introduces successful steps for both learning and teaching process. Communication is more than words, and it is important for teacher to understand the non-verbal message they are sending and receiving in the classroom. Non-verbal message includes facial expressions, eye contact or lack of eye contact proximity and closeness, hand gestures, and body language. Much of the research about non-verbal communication indicates as little as 7% of communication is spoken words and maturity is non-verbal and paralinguistic cause. Hence, it is critical for teachers to learn to apply non-verbal communication signals in the classroom. Apart from the theoretical courses and references that help novice teachers to passively learn about teaching proficiency basic such as communication and management skills, simulation based system training provide a safe and comfortable environment for them to interactively practice teaching skills in our realistic classroom.

Ranked sixth, *well groomed and well-poised with weighted mean 2.37* interpreted *affected much. In fact, proper grooming takes the spotlight away* from their self. The students pay attention on what you are saying and really get to know you as the teacher. Personal grooming and well-poised is about presenting yourself in the best possible way. It will help develop a wholesome

such as changing habits, attitudes, reconstituting ideals and personality interest of the learner. To help the students to acquire, retain and be able to use knowledge. Under stand, analyse, synthesize and evaluate the knowledge and achieve skills, establish habits, and development attitudes. Teaching is the major occupation of man for which has not learners personal judgment are essential and significant factors in performing well. pleasant teachers are well groomed, neat, presentable and well-poised. Similarly, attractive and Teacher who understands this precept and adheres to it

enlarge the importance of his/her dress and grooming. the Academic Performance of Bachelor

c) Classroom Setting Factor Affecting

of Secondary Education. the degree of effect of the classroom setting

Table 5 below shows the Bachelor of Secondary Education performances of factors in the learning

ent or classroom ecology refers to the general and overall in College Algebra.

of a classroom. Teachers create, provide, or facilitate Classroom environment

orient for learning. Effective classroom environments of climate and atmosphere

be substantially different from those of the the classroom environment

the 21st-century Will ultimately Factors Affecting the Academic Performances of

Setting in College Algebra		WM	DE	
1. Classroom	Students	2.73	AMu	1.5
2. Classroom	Seating	2.32	AMo	5
3. Classroom	Ventilation	2.34	AMo	4
4. Classroom	Seating Arrangement	2.73	AMu	1.5
5. Classroom	Classmates	2.67	AMu	3
6. Classroom	Cleanliness		2.49	AMo
7. Classroom	Capacity			
8. Classroom	Room			
9. Classroom	Classroom			

Average mean

a broad range of industrial era. Classroom environment encompasses physical setting, the psychological educational concepts, including the physical and numerous instructional environment created through social contexts, teacher characteristics and behaviors. The study components related to classroom environment has been psychology. Researchers are interested in specializations of educational performances of the students in relationships between environment and the physical environment. The mid-1990s has focused on one or more of these aspects and student outcomes. More frequently a with numerous positive and negative environment, the physical environment focus in earlier studies of classroom studies as an influence on has continued to appear in contemporary studies of the physical behavioral and academic outcomes. Current studies of the physical environment have investigated the physical classroom setting. Classroom setting has a great impact also in the student's or capacity, class size, a factor ventilation and cleanliness with an average Classroom setting much the learning performances of the achievement. Under the findings, conducted survey described in the of 213 respondents. Accord to these, 88 percent of the students saying they are respondents. cation that 88 percent of the students saying they are Chronicle of Higher Education in even surveyed said good learning distracted from learning of the student's study. If the students' feels uncomfortable Furthermore, DA perGent t If should be spotless environment

of the classroom condition, then they will have less concentration on the lesson and so they get a little information only, from the teacher, and thus affecting their performances. Temperature and ventilation system inside the classroom learning environment. re crucial factors that affects classroom a perform significantly better on classrooms Students in well-ventilated classrooms who receive inadequately fresh air, standardized tests than their peers according to newly published research conducted by The University many schools fall short of providing a healthy Indoor Air Program. "Far too By doing something asy for children. simple help learning environment into the classroom, schools potential! could program introducing more fresh air every student perform at a higher level," said Richard Shaughnessy, p Research at TU and a research associate in the that is too cold or too hot director of Indoor Air Quality Engineering. Classroom i performances and concentration as they feel Department of Chemical student's According to Halstead (1974), it is generally negatively affects the conditions. creates physiological and uncomfortable in such temperature and humidity fatigue, causes people to work more accepted that high which expedite to make more mistake and errors. psychological problems h efforts and is central to student's academic slowly, apply mu°. and ventilation managed not only to provide physical Thus, cleanliness e caution sill factor in the learning process by achievement. It s- cerVe as a positive d oncentration. comfort but also "- an attentiveness stimulating

Classroom capacity ranked third with a weighted mean of 2.67 affects moderately the learning performances of the respondent. The Performances of the Bachelor of Secondary Education is moderately affected by the classroom capacity. Studies about class size have examined how class size influences smaller classes are associated with student and teacher behaviors. In general, students are more frequently on-task with fewer reported behavior problems than students in larger classes. Although teachers use similar instructional strategies whether teaching large or small classes, there is evidence to suggest that more class time is spent on administrative tasks for larger classes, leaving less time available for instruction. Some research has suggested that differences in academic outcomes based on that overcrowded facilities, too many students in classrooms are three major issues cited as cited by Miller (2011) to increased stress levels of students and classes and lack of teachers' due to behavioral problems. These potentially increased incidences of behavior problems found in larger classrooms are increased teacher-reported and academic achievement. Increased stress levels are by lower levels of interest to many researchers frequently accompanied by ratios are also of behavioral problems seems to off. Teachers-to-child reported have observed that large classes, with a number of students more because the number of students increases. Moreover, 30 or more students, tend to have a larger number of students with the teacher than children in small classes. With fewer students engaged often with

have students with *more positive perceptions* of fairness in grading, stronger class cohesion, and higher *degree* of social support, as well as higher achievement scores. Female students have been found to prefer collaborating with other students when studying and resolving problems and they have a stronger preference for teacher support than male students. The primary school *environments* tend to use collaborative strategies more frequently and have higher levels of teacher involvement and support than is found in secondary schools. Research on single-sex classrooms has been more divided *in terms of academic outcome research*. Some studies found that girls do better *in* math and science particularly when separated from male students; other studies found no achievement differences between genders when either in *single-sex* or *mixed-sex* classrooms. The interactions among students in the classroom are a *normal and* essential part of the learning process that influences the lifelong *learning* habits of *students*. The potential effects of classmate or peer relationships are *reciprocal*: Some students are more receptive than others. On one extreme, *for* example, is the student who values and seeks peer input *on* every decision; on the other is the social isolate who avoids interaction in *and out* of the classroom. This entry examines selected *variables that can influence* learners, including developmental differences, *note* and *learn^{ing}* considerations, and the function of the classroom contexts. In a 1953 book, Henry Stack Sullivan outlined a developmental theory *describ^{ing} the chang^{es} in* inter-personal *nee^{ds}* as an individual matures they can *cope to different* kinds of personalities around them. Thus,

classmates has moderately affects the academic performance of one individual in certain ways *but can* be controlled by a learner not to *hinder* him in performing best *in learning*.

Fifth is the seating arrangement with an average weighted mean of 2.32 interpreted to be *affecting* moderately the performance of the respondents. The Bachelor of Secondary Education student thinks that they can perform in mathematics *no matter* who their seatmates are. It is indeed true that their seatmates doesn't define how will one individual perform. In our country, it is typical that there is a seating arrangement in every classroom it is one of the rules inside *the school*. Seating arrangement may vary in different approaches arranged by the teacher. It is commonly base on *their* behaviors inside the classroom.

Summary

Based *on the foregoing* findings the study revealed that:

- 1) The level of performance of Bachelor of Secondary Education students in College *Algebra* is satisfactory with a weighted mean of 80.3.
- 2) The Identified factors much affect the academic performance of the students in College *Algebra* with a weighted mean of 2.63.

CONCLUSIONS AND RECOMMENDATIONS

This part of the study presents the conclusions, and recommendations based on the findings of the study.

Conclusions

Based on the foregoing findings, the following conclusions were drawn:

- 1) The level of performances of *Bachelor* of Secondary Education juniors students in *College Algebra* is average.
- 2) The identified factors to academic *performance* of the students in *College Algebra* have *much effect*.

Recommendations

Based on the *foregoing* findings and conclusions, the following recommendations are offered:

- 1) Mathematical organizations should be established to maintain and provide activities *in line* with mathematics that will further develop the mathematical ability of the students. An organization that embark the importance of *teaming* mathematics in every individuals endeavor.
- 2) The top factors affecting the performances of • the Bachelor of Secondary education *junior's* stated in the study must be given priority by the *instructors and students* in the teaching-learning process to maintain *and balance the* effects of the factors or to eliminate it.

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- 3) Additional mathematics books and materials should be considered in the library accessible to the student to enhance the performances of the students in Mathematics.
 - 4) Quiz bees and competitions within the campus wherein rewards such as certification will be given.
 - 5) Further studies in the learning performances of Education students should be conducted. Specifically in the alternative method of teaching mathematics

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APPENDECES

APPENDIX A



KING'S COLLEGE OF THE PHILIPPINES
Pico, La Trinidad, Benguet
COLLEGE OF TEACHER EDUCATION

Dr. Jovita Fernandez
The Registrar

October 27, 2015

Dear Ma'am,

The undersigned are conducting a research entitled "ACADEMIC PERFORMANCE OF BACHELOR IN SECONDARY EDUCATION STUDENTS IN COLLEGE ALGEBRA" in partial fulfillment in Action research in Mathematics (Math 422).

In this regard we would like to acquire the grade of the currently enrolled third year students when they took College Algebra.

Rest assured that the data collected will be kept confidential and solely be used for the purpose of study.

We are hoping for your favourable response on this request.

Truly yours,

FEJ Y R4GUITAN

CLAIRE M. MARCOS

JENNY P. KIMBES

IARE rH (5, 9L-IWEN

Noted by:

CHR/STINE FLOR B. WANSI
Aarvis6r

RUFINA D. CADSI, PhD.
Dean

APPENDIX B



KING'S COLLEGE OF THE PHILIPPINES
Pico, La Trinidad, Benguet
COLLEGE OF TEACHER EDUCATION

October 26,2015

Ms.Rufina D. Cadsì,PhD
DEAN, College of teacher Education
King's College of the Philippines

Ma'am,

The undersigned are conducting a research entitled "**ACADEMIC PERFORMANCE OF BACHELOR IN SECONDARY EDUCATION STUDENTS IN COLLEGE ALGEBRA**" in partial fulfilment in Action research in Mathematics (Math 422).

In this regard we would like to seek permission from your good office to administer our questionnaire to all Third year Secondary Education Students of King's College of the Philippines as they are the subject of our research.

Rest assured that the data collected will be kept confidential and solely be used for the purpose of study.

We are hoping for your favourable response on this request.

Truly yours,

FEI Y. BAGUITAN

CLAIRE M. MARCOS

JENNY P. KIMBES

JARETH G. PAL-IWEN

Approved by:

RUFINA D.CADSI, PhD
College Dean

APPENDIX C



KING'S COLLEGE OF THE PHILIPPINES
Pico, La Trinidad, Benguet
COLLEGE OF TEACHER EDUCATION

Dear respondent,

The undersigned are conducting a study entitled "ACADEMIC PERFORMANCE OF BACHELOR IN SECONDARY EDUCATION STUDENTS IN COLLEGE ALGEBRA" as a partial fulfillment of the requirement in Action Research ⁱⁿ Mathematics (Math 422).

In this regard we have chosen you to be our respondent of this study and we hope that you will take *time answering* the questions honestly.

Rest assured that the data gathered will be treated with utmost confidentiality and will be used for academic purposes only.

Thank you for your kind cooperation.

Respectfully yours,

Fei Y. Baguitan

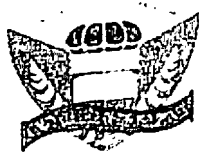
Jenny P. Kimbes

Claire M. Marcos

Jareth G. Pal-iwen

APPENDIX D

KING'S COLLEGE OF THE PHILIPPINES
Pico, La Trinidad, Benguet
COLLEGE OF TEACHER EDUCATION



Name:

Gender:

Major:

Put a checkmark on the column that describes the factors as they affect your learning. Be guided by the following legend:

4 —Affects very much
3 —affects moderately
2 —affects slightly
1 —less affect

3 —affects much
1 —less affect

	4	3	2	1
Student Factor				
1. mathematical ability				
2. learning style				
3. Economic status				
4. Health condition				
5. attitude				
6. number of units enrolled				
Teacher Factor				
1. Well-roomed and well-ventilated				
2. Verbal and nonverbal communication				
3. Choice of instructional materials				
4. Clarity of voice				
5. teaching style				
6. mastery of subject matter				
Classroom setting factor				
1. Ventilation				
2. Seating arrangement				
3. Classmates				
4. Cleanliness				
5. Classroom capacity				

Biographical Sketches

Author A

Fei Yap-eo Baguitan was born on February 16, 1995 at Dalip-ac, Balili, Mankayan, Benguet. She was the last daughter among the 3 siblings of Mr. Pedro Baguitan and Mrs. Elvamina Baguitan. She believes in the sayings that Education is the Key to Success. That in every hardship that she encounters in her studies she couldn't easily give, but rather she is still be firm and do all what she can. In every troubles she is experiencing she use it as her challenges in life to be successful in all the things she do.

She Graduated at Mogao Elementary school at Mogao, Balili, Mankayan, Benguet in the year of 2007-2008. She graduated at Bay-o, National High school at Bay-o, Balili, Mankayan, Benguet in the year of 2011-2012. As she graduated she also receives some special awards. So with the love and help of the almighty God she is now having her studies at King's College of the Philippines taking up the Bachelor of Secondary Education major in English a 4th year BSED students.

She is very thankful to the Almighty God for guiding her, giving wisdom, knowledge and strength in all she does in her studies. She is also proud to her family in providing all her needs. And by giving her moral supports in which they are not tired in giving her advice. And to her friends by always there to accom pany her in all th^e thing^s she does in the schools. They are there to cheer him in times she cannot do her task in the school as well as to the other peo ple around her by giving respect, love, value and acceptance.

Author C

Claire Villanueva Marcos was the second child among the five siblings of Mr. Agustin and Siblyn Marcos is currently enrolled as a fourth year student at King's College of the Philippines taking up Bachelor of Secondary education. She was motivated by the supports given by her majoring in Mathematics. More on her studies. She believes that to parents and relatives to pursue to succeed in life dare to be different.

She finished her Elementary years at Patiacan Elementary School and being a young dreamer she left her beloved mountainous home to continue her studies. Motivated by her ideals and dreams in life she was able to surpass the challenge of being a student and was able to graduate at 2012. She loves watching fairytales to school in the year. She was fascinated by the beauty of Quirino National High situations. She was fascinated by the beauty of and movies based on real life. Reading books was her nature which drives her for joining outdoor activities. With the past time in the province, relatives, friends and of course the loving God undying help of her parents, relatives, friends and circumstances which she whom she trusts the most in times of troubles and circumstances which she she was able to continue her dream. Still she was able to fight and win in every battle in this have no control and optimistic girl who hopes in a changing World. constant

Author D

Jareth G. Paliwen was born on **January 17, 1996** at **Am-am, Balili, Mankayan, Benguet**. *She was the fifth daughter* of **Mr. Candido Pal-iwen** and **Mrs. Petra Pal-iwen**.

She finished her elementary years at **Am-am, Elementary School Am-am, Balili, Mankayan, Benguet** *in the year of 2007-2008*. **She graduated High school at Am-am, National High School-Am-am, Cagubatan, Tadian, Mountain Province** *in the year of 2011-2012*. **From first year to fourth year she was a player of badminton during intramurals**. *She experience to attend district meet during third year and fourth year but she is not lucky to go in provincial meet*. *During third year and fourth year high school she got some awards*. **She loved to play guitar, to hear Christian songs, to watch anime and action movies**. **She loved solving most** *if it has formulas even though it is hard*. **With the support of her mother she was able to continue college**. **At present she is 4th year student studying at Kings College of the Philippines taking the course of Bachelor of Secondary Education** *majoring in Math*. **Taking Math as her major is very challenging and full of pressure**. *Through the help of God she is able to finish and face every challenge in this course that she took U.* **She is acknowledging that God is her courage and the one who gives confidence to her.**

Her favourite verse in the bible is Philippians 4:13, **can do all things through Christ who gives me strength.**